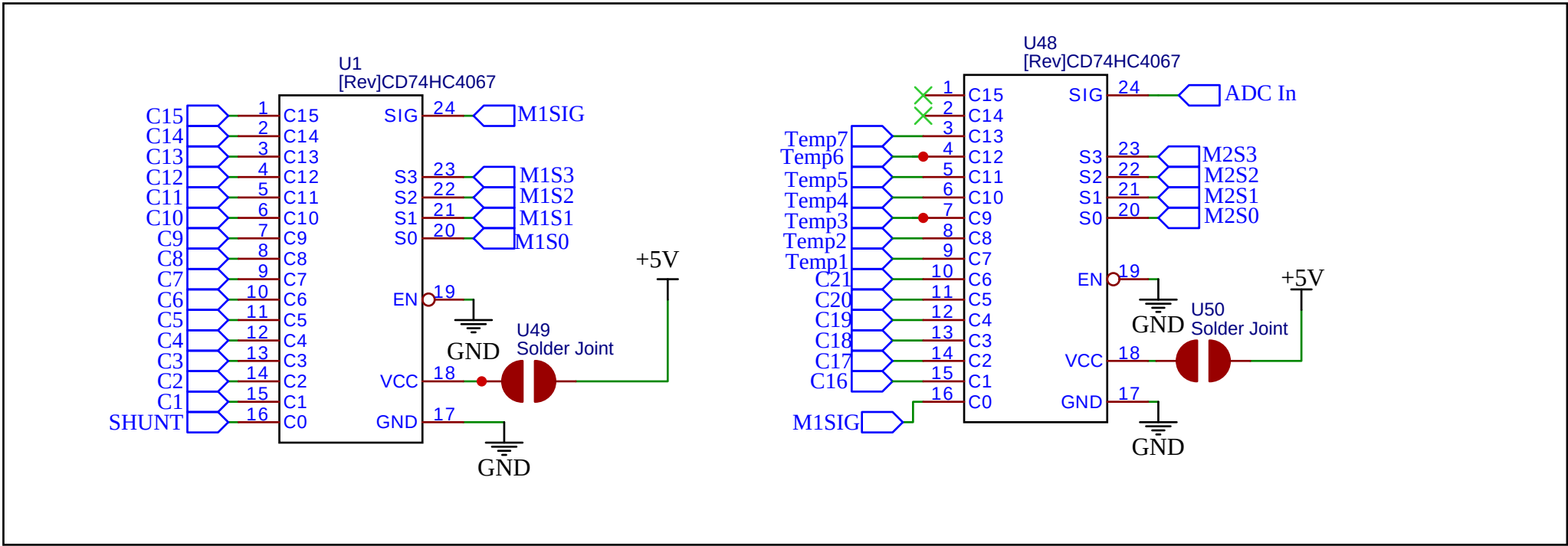
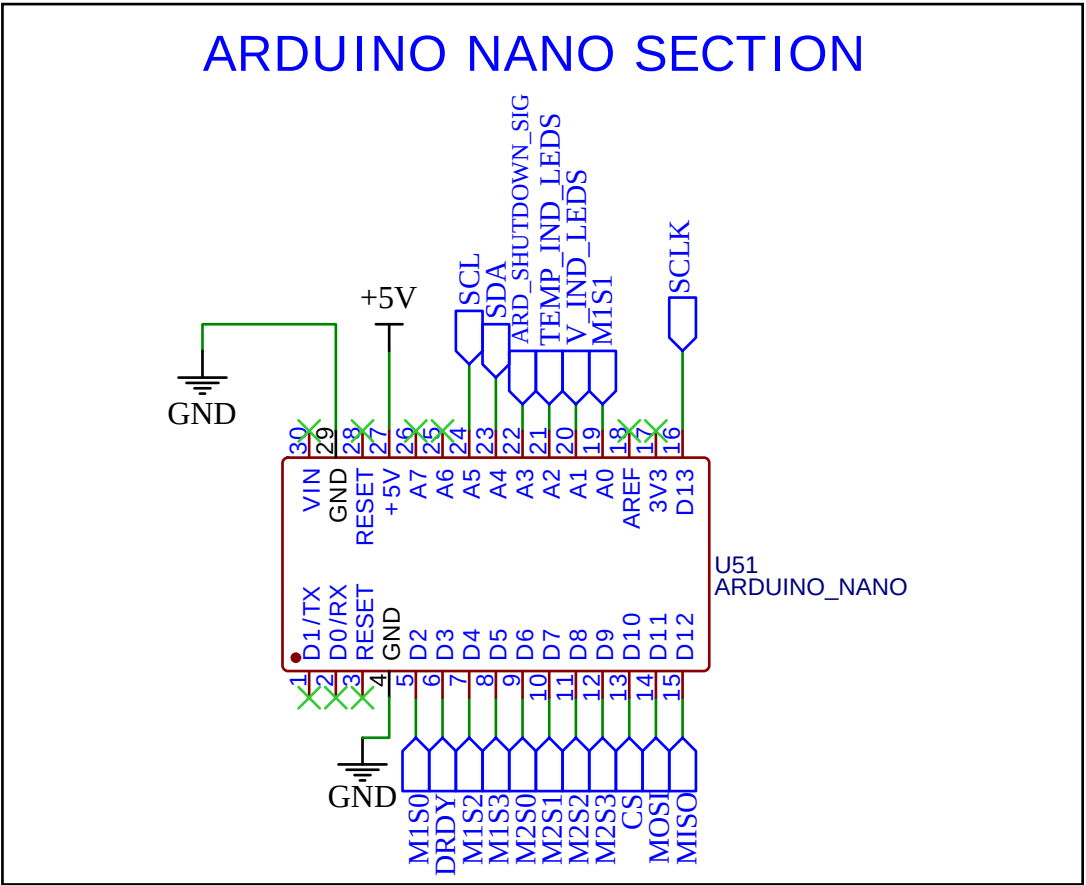


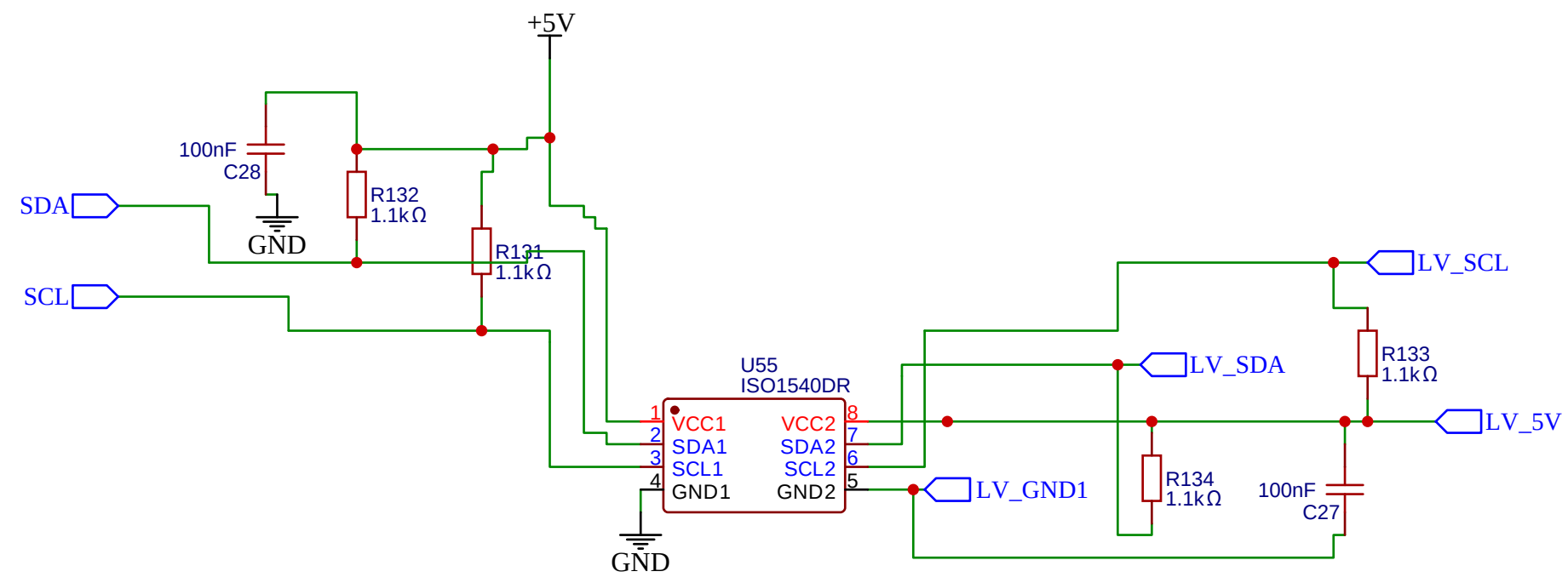
MUX SECTION



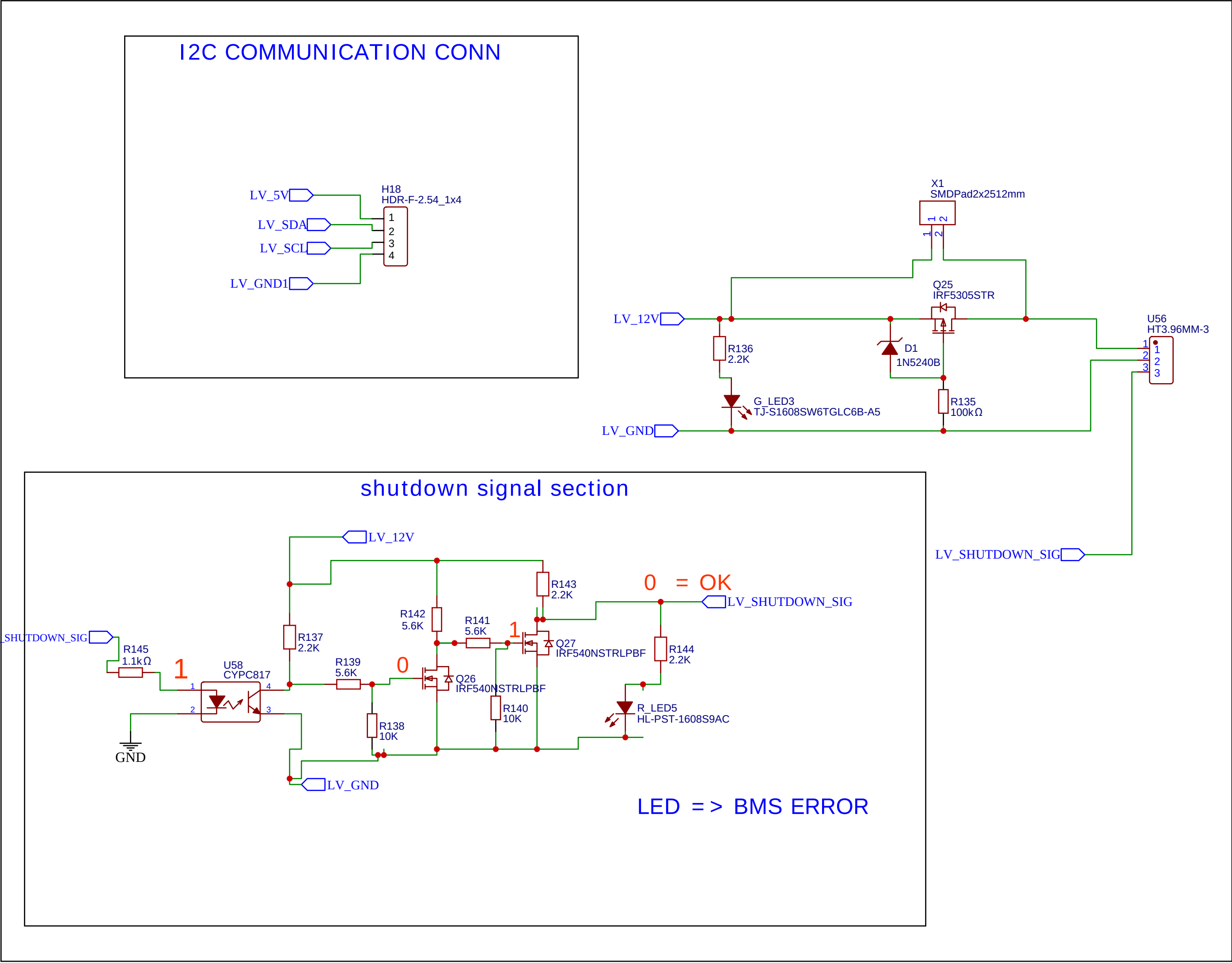
ARDUINO NANO SECTION



I2C COMMUNICATION SECTION



TITLE: I2C COMMUNICATION SECTION		REV: 1.0
	Company: FERN (BMS)	Sheet: 5/8
	Date: 2023-01-29	Drawn By: M. Hammad Hassan Mallick



TITLE: LV SECTION		REV: 1.0
	Company: FERN (BMS)	Sheet: 6/8
	Date: 2023-01-29	Drawn By: M. Hammad Hassan Mallick

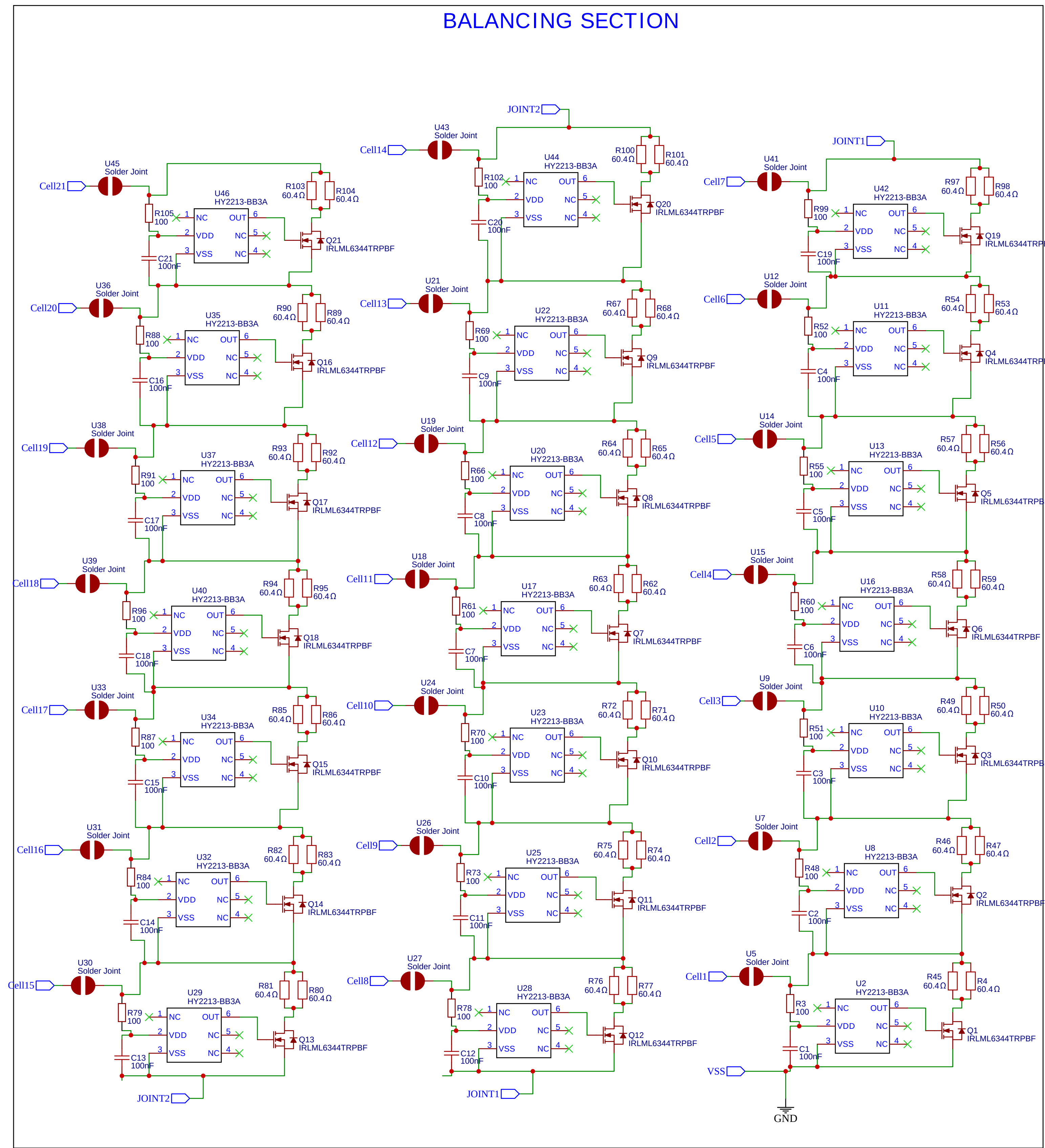
temp v dividers

Diagram illustrating seven temperature sensor voltage dividers (temp1 to temp7) connected to ground (GND).

The components and connections for each divider are:

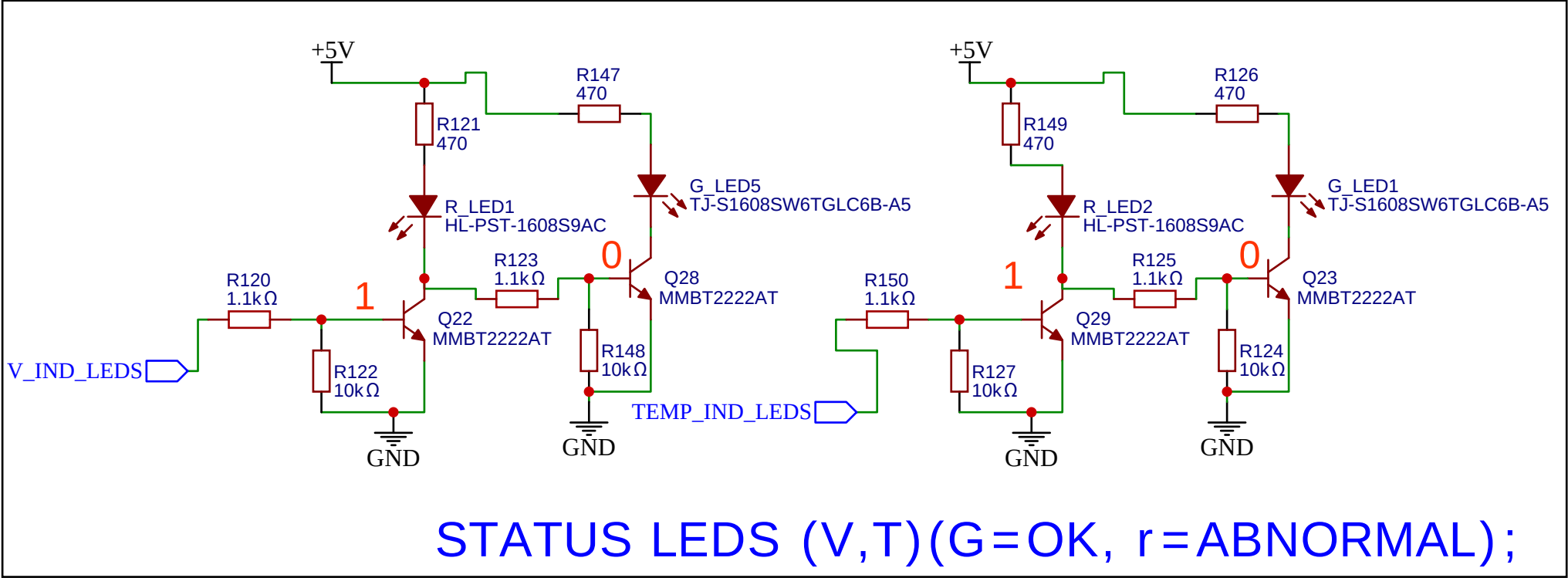
- temp1:** Sensor t1 connected to R106 (5.6K), which is connected to R107 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.
- temp2:** Sensor t2 connected to R111 (5.6K), which is connected to R110 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.
- temp3:** Sensor t3 connected to R115 (5.6K), which is connected to R114 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.
- temp4:** Sensor t4 connected to R119 (5.6K), which is connected to R118 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.
- temp5:** Sensor t5 connected to R109 (5.6K), which is connected to R108 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.
- temp6:** Sensor t6 connected to R113 (5.6K), which is connected to R112 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.
- temp7:** Sensor t7 connected to R117 (5.6K), which is connected to R116 (5.6K). The junction is connected to GND. The sensor output is connected to the junction point.

BALANCING SECTION



NOTE:

IF CELL VOLTAGE INCREASES ABOVE 4.2
IT WILL START DISCHARGING THROUGH RESISTOR
IC USED HAS INTERNAL REFERENCE FOR COMPARING FOR 4.2.

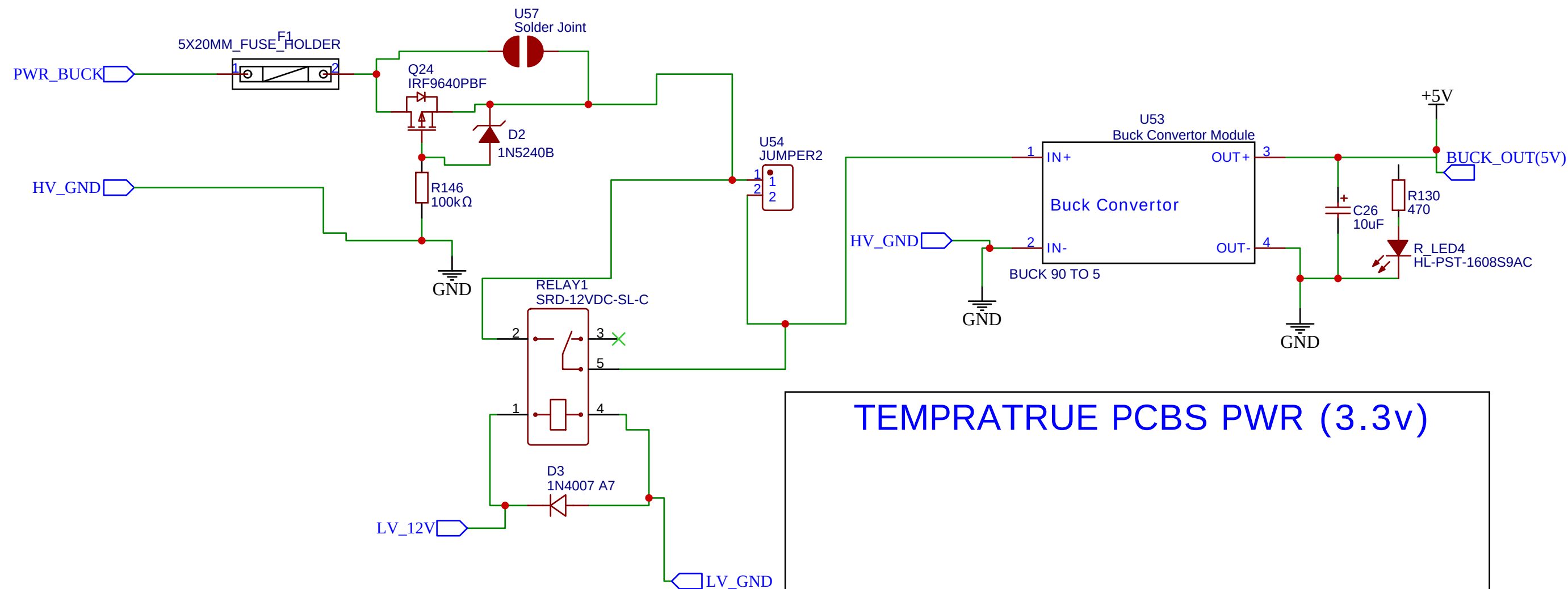


NOTE:

GREEN LEDS WILL LIGHT UP IF TEMPRATURE AND VOLTAGE IS NORMAL (OK STATE)

RED LEDS WILL LIGHT UP IF TEMPRATURE AND VOLTAGE ARE EITHER HIGH OR TOO LOW (ERROR STATE)

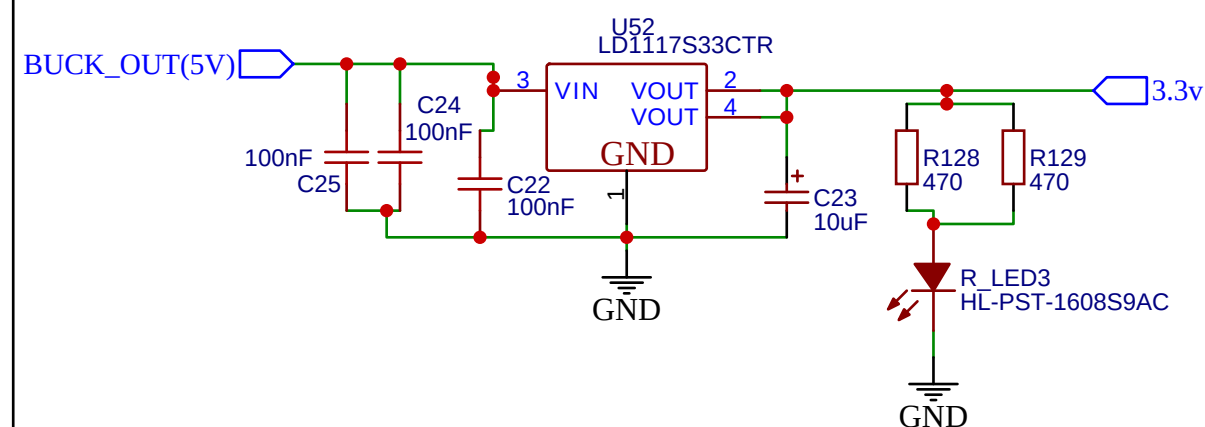
TITLE: INDICATION LEDS		REV: 1.0
	Company: FERN (BMS)	Sheet: 7/8
	Date: 2023-01-29	Drawn By: M. Hammad Hassan Mallick



NOTE:

- 1: LV driven relay switch HV to buck input
- 2: fuse rating to be selected
- 3: reverse polarity protection , can be bypassed
- 4: 3.3v used for powering up TEMP PCBs (mux issue)

TEMPRATRUE PCBS PWR (3.3v)



TITLE: POWER INPUT SECTION		REV: 1.0
嘉立创EDA	Company: FERN (BMS)	Sheet: 1/8
	Date: 2023-01-29	Drawn By: M. Hammad Hassan Mallick